HEATMAP NODE · RISK REGISTER AND HAZARD ANALYSIS

HeatMap Node risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-01	Amish Chadha	First register at the kit 1.9.1 upgrade: project risks, preliminary hazard analysis and design FMEA for options A and B

HeatMap Node risk register and hazard analysis

The biggest risks are that the derived WBGT does not match a reference instrument on a street (PR-005), that a head or globe falls from a pole in wind, and that nobody keeps the nodes up. The hazards that matter most to people are a fall from a ladder at a pole and contact with an energized street light pole; both are controlled by procedure, not by this design alone. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high).

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R1	Derived WBGT is off in streets	Liljegren model not validated for street canyons; calm intervals	Maps misguide siting (PR-005 at risk)	3	4	Design lead	Collocate with a reference WBGT meter at TRL 4; consider option B reference heads (HMN-OPT-001)
R2	No partner pole	Cities slow to grant attachment	No trial network	3	4	Amish	Approach the first candidate city after Amish agrees; nothing is agreed (HMN-DEC-001)
R3	Cost grows beyond the target	Prices change; the head is already USD 6 over	Network too costly	4	3	Design lead	Savings in HMN-DEC-001; Amish decides any change to the target
R4	Sensors at 2.8 m misrepresent pedestrians	Height chosen for reach and vandal resistance	Bias of 0.2 to 0.8 °C in strong sun (PR-009)	4	2	Design lead	Report the correction; test 2.0 m against 2.8 m (HMN-DEC-001 O2)
R5	FieldNode changes under the head	Shared core is updated in its own repo	Interface drift (adapter, ports)	3	3	Design lead	Interfaces listed in HMN-DEC-001, to confirm 7; cross-repo note in REVIEW.md
R6	Upkeep not kept	Volunteers, no funding	Fouled shields, stalled fans	3	3	Partner	Visit interval and steps in the plan (PR-018); option A chosen partly for this reason
R7	Theft or vandalism	Visible head at 2.8 m	Loss of node	2	3	Partner	Height, stainless bands that need a tool to undo, no scrap value (aluminium and ASA only)
R8	Data misread as a heat warning	Maps shared without context	Wrong action by the public	2	4	Amish	State that the node supplies data and public warnings stay with the responsible agency (HMN-PRB-001, Out of scope)
R9	Cell or electronics fail in heat	57.2 °C interior at 50 °C air	Node offline; cell damage	3	4	FieldNode lead	Charge cut-off (PR-016, STEM S25)

Preliminary hazard analysis

Table 2. Hazards.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Fall from a ladder or lift at 2.8 m	Fitting the arm on a street pole	Serious injury	5	3	Two people; lift or secured ladder; fit only where the owner permits; plan notes (HMN-BLD-001)	3
H2	Contact with an energized pole	Street light pole faulted to ground	Electric shock	5	1	Only poles the owner has isolated or checks; owner's technician; no work on a wet pole	2
H3	A part falls on a passer-by	Band slip, boss failure, wind	Head injury	5	2	Lanyards on the globe and the shield; clamp twist factor 2.5; 35 m/s case (HMN-CAL-001); check at the first visit	3
H4	Lithium cell fault	Charging in the heat; crushing; short	Fire, venting	5	2	LiFePO4 chemistry; charge cut-off (O4); cell and BMS in FieldNode's own safety case	3
H5	Burns from hot parts	Globe up to 67 °C and shield plates 65 to 75 °C in the sun	Minor burn to a person fitting or servicing	2	3	Wear gloves; let parts cool; labels	1
H6	Cuts and pinches	Formed aluminium edges, strap bands, fan	Cuts	2	3	Edges broken (cad_check edge breaks); finger guard on the fan; gloves	1
H7	Fan entanglement	Cable or finger in a 60 mm fan	Minor injury	2	2	Cowl with legs over the fan; fan stops between readings	1
H8	Manual handling	Carrying a 4 to 5 kg node up a ladder	Strain	2	3	Lift the core and the arm separately; hoist rope	1
H9	Mistaken for a surveillance device	A box on a pole	Public concern	2	3	Label the node; no camera, microphone or radio scan (PR-012); publish the data format	1
H10	Standing water in the reservoir (option B only)	A closed bottle with a 6 mm gland	Mosquito or biofilm growth	2	2	Closed bottle; inspect at refill (STEM S36)	1

Design FMEA (first pass, P3)

Table 3. Failure modes, sorted by RPN. RPN = severity x occurrence x detection. F11 and F12 are option B only; all others are option A (F1 to F10, F13 to F15), and most also hold for B.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Shield plates and gaps	Pass air over the sensor	Gaps blocked by insects, dust or a nest	Air temperature reads high in the sun	Sheltered 12 mm gaps; cowl lid	4	4	4	64	Inspect gaps at each visit (PR-018); screened inlet to consider (STEM S27)
F13	Aspiration fan	Keep the shield ventilated	Fan stops or slows without any signal	Air temperature reads 1 to 2 C high in sun, unnoticed	Dust, bearing wear, about 1.75 M starts in 10 years	4	3	4	48	Add a tach or airflow check to the sensor step response; flag stuck readings (independent review)
F14	Black globe	Read radiant heat	Rain or dew cools the globe by evaporation; a bird perches on it; dent in the 0.4 mm shell	Globe temperature reads low or biased	Weather, wildlife	3	4	4	48	Flag rain from the humidity channel; spikes on the globe top to consider
F8	FieldNode cell	Store energy for 5.75 days	Charges above its rated temperature in the 57.2 °C interior	Capacity fades; at worst a cell fire	Sun load at 50 °C air	5	2	4	40	Cut charging above the cell's limit in FieldNode firmware (STEM S25, PR-016, HMN-DEC-001 O4)
F15	Arm and clamp	Carry the head in wind	Fatigue crack at the bolt holes at the arm root	Arm drops or twists	Wind-induced vibration over years	5	2	4	40	Bolt holes at the root to check at TRL 4 (STEM S9); inspect at each visit
F2	Aspiration fan	Draw 4.0 L/s through the stack	Bearing wears or stalls	Shield error rises to 1.04 °C at 1 m/s; reading biased in calm sun	Dust, heat, 6 s run every 3 min	4	3	3	36	Flag a fan fault by the sensor's own step response; spare fan; fan curve to measure (S7)
F5	Globe probe	Measure globe temperature	Bead drifts or leaks moisture	Globe reading biased	Epoxy ageing, moisture in the boss	3	3	4	36	Recalibrate in CalRig yearly; sealed boss
F12	Wick (option B)	Carry water to the sensor	Dust and salts block it	Wet bulb reads high		3	4	3	36	Distilled water; change the wick

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
					Street dust; hard water					at each visit (S35)
F3	Globe and hanger tube	Hold the 150 mm black globe	Hanger tube or boss fails; globe falls	Globe lost; falls on the street	Wind and fatigue of the soldered boss; vibration	5	2	3	30	Secondary lanyard (line 11); check the boss at each visit (HMN-DEC-001, to confirm 1)
F4	Clamp saddle and strap bands	Hold the arm to the pole	Band slips or relaxes; arm twists	Readings face the wrong way; at worst the head falls	Preload below 1,000 N; rubber trim creep	5	2	3	30	Check the preload with the band tool (HMN-DEC-001, to confirm 6); lanyard to the pole; re-tighten at the first visit
F6	SHT45 membrane cap	Admit air, keep out water	Cap clogged with dust	Humidity lags and reads wrong	Dust and salt	3	3	3	27	Replace the cap at each visit; reference reading from option B head where fitted
F7	Cup anemometer	Measure wind	Bearing stalls or start-up above 0.8 m/s	Calm intervals misread as zero	Wear, dust, ice	3	3	3	27	Calm share flagged per interval (PR-004); replace at 2 years
F9	M12 leads and plugs	Connect sensors to FieldNode	Water enters a plug; UV cracks the jacket	Channel lost	Loose plug; sun	3	3	3	27	IP67 plugs; drip loops; cable ties
F10	Printed ASA parts	Hold plates and the cowl	Cracks or warps at 65 to 75 °C	Plate gaps close; cowl sags	Filament with low heat deflection	3	3	3	27	Specify the filament's heat deflection above 90 °C (S23); inspect at each visit
F11	Reservoir bottle (option B)	Hold water for the wick	Freezes and splits; runs dry	No wet-bulb reading; water leak	Brim fill; hot dry spell	3	4	2	24	85 % fill line and a channel flag (K1); 5 day refill in hot spells (K1b)

Scores are the author's judgement at TRL 3 and have not been reviewed with a co-design partner. Items at 40 or above come first in the field comparison.

Sources

- ISO 7243:2017, https://www.iso.org/standard/67188.html, accessed 2026-11-01 (hazard context).
- HMN-STM-001 rows S7, S11, S22 to S27, S31 to S36 and HMN-CAL-001 sections H and F (design evidence).